KEATON B. SHURILLA

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Education

University of Tokyo, Tokyo, Japan **Oct 2026 – Present**
PhD in Advanced Interdisciplinary Studies
Advisor: Prof. Masahiro Nomura

University of Arizona, Tucson AZ, USA **Aug 2026**
MS in Optical Sciences
Thesis: *Multi-Wavelength Reconstruction in Volume Holography: Fabrication of 1550 nm Gratings with Visible Light*
Advisor: Prof. Yuzuru Takashima

University of Tsukuba, Ibaraki-ken, Japan **Mar 2024**
MS in Informatics
Thesis: *Audiovisual Plasma Volumetric Display: Effect of Spatial Modulation on Optoacoustic Transduction*
Advisor: Prof. Yoichi Ochiai

Brigham Young University, Provo UT, USA **June 2021**
BS in Electrical Engineering, with university honors
Thesis: *Sonic Glass: Display Compatible Aerial Tactile Haptics with Annular PMUTS*
Advisor: Prof. Daniel Smalley

Professional & Research Experience

Nomura Laboratory, University of Tokyo | Graduate Research Assistant **Oct 2026 – Present**
Link: https://nlab.iis.u-tokyo.ac.jp/

Extensive FEM simulation and cleanroom fabrication of acousto-optic and phononic devices. JST SPRING GX project student.

Advanced LiDAR and Display Lab, University of Arizona | Graduate Research Assistant **June 2025 – Aug 2026**
Link: https://wp.optics.arizona.edu/takashima/

Development of flat holographic optics for free space optical communications and display. Fabrication of infrared wavelength holograms exposed using visible wavelengths. Beam steering with MEMS phase light modulators. Finite element simulations of MEMS based beam steering systems and grating waveguide couplers.

Quantum NanophoXonics Lab, University of Arizona | Graduate Research Assistant **Aug 2024 – June 2025**
Link: https://wp.optics.arizona.edu/eichenfield/

Finite element simulation of acousto-optic and optomechanical photonic devices. Experimental characterization of acoustoelectric and photonic resonator devices at telecom frequencies. Development of tapered and dimpled fiber couplers. Invention disclosure to the University of Arizona tech transfer office on acoustoelectric methods for enhancing acousto-optic deflectors.

Digital Nature Group, University of Tsukuba | Graduate Research Assistant **Oct 2021 – June 2024**
Link: https://digitalnature.slis.tsukuba.ac.jp/

R&D on polymer film-based holography and HOE fabrication. R&D on integrated photonics for holography. Fabrication of diffused waveguides in lithium niobate.

Successful development of femto-second laser-based plasma audio system.
Development of fluorescent particle based optical trap volumetric display.

CYBERDYNE INC. | Part-time Research Assistant **Aug 2023 – June 2024**
Link: https://www.cyberdyne.jp/

R&D on LED based photoacoustic imaging system for low-cost health care devices. Developed an optical power and thermal measurement system using automated scripting and computer vision. LED array metrology.

BYU Electro-Holography Lab, Provo, | Undergraduate Research Assistant **Feb 2019 – June 2021**
Link: https://www.smalleyholography.org/

Fabrication of transparent piezoelectric micromachined ultrasonic transducers (PMUT) in a phased array configuration for the purpose of generating tactile feedback. Various holography and volumetric display projects. Extensive work on transducer design, UV photolithography and other cleanroom fabrication techniques. Additional work on PCB design and extensive research into piezoelectric polymers, phased arrays, and other novel tactile feedback projects. Regular public presentations about my research to other students and faculty.

iProspect Japan, Tokyo, Japan | Assistant to the APAC Head of Innovation **April 2018 – Sep 2018**

Internship directly under the APAC head of innovation and e-commerce at the Tokyo office of iProspect, part of the Dentsu-Aegis network. Researched current trends in marketing tech, with a specific focus on voice assistants, IoT, and VR/AR technologies. Some programming design work for Google Home and Amazon Alexa. Additionally, I took part in AdWeek Asia and was invited to represent our company at the AWS Developers Summit. Lastly, I also did some translation for others in the office.

BYU Electrical Engineering Shop, Provo, UT | Technical Advisor **Jan 2017 – April 2019**

Advisor at the campus makerspace. Assisted students with personal projects and instructed them on the proper use of electrical, woodworking, and metalworking equipment. Use of 3D printers, PCB milling machines, and laser-cutters on behalf of students and professors. Designed test rigs for various engineering courses and research labs. Oversaw assembly for the kits used in different classes. Management of an inventory of thousands of electronics components.

Honors and Awards

Spring GX Scholarship, Japan Science and Technology Agency (JST) **2026**

Waitlist - National Defense Science and Engineering Graduate (NDSEG) Fellowship **2025**

Joseph W. Goodman Graduate Student Endowed Scholarship in Optical Sciences **2024**

1st Alternate - DC1 Doctoral Fellowship, Japan Society for the Promotion of Science **2023**

1st Place - IEEE AP-S International Student Design Contest **July 2021**

NASA Space Grant - NASA/STEM Micro-Museums: Rural 'Science Satellites' for Space Education **2019**

Publications

Google Scholar: link

5. **Keaton Shurilla**, Jeffrey Y. Chen, Tianyao Zhang, et al. "Fabricating telecom wavelength volume holograms using visible light for infrared applications", Proc. SPIE 14222, ODS 2026: Industrial Optical Devices and Systems, 142220G (8 Oct 2026); https://doi.org/10.1117/12.3113692

4. Jeffrey Y. Chen, Muralidhar M. Balaji, Gregory M. Nero, John M. Bass, Ignacio A. Rojas, Masahiko Yasui, **Keaton Shurilla**, et al. "Pointing, acquisition, and tracking at 1550nm using MEMS phase light modulators", Proc. SPIE 14227, Laser Communication and Propagation through the Atmosphere and Oceans XV, 142270A (9 Oct 2026); https://doi.org/10.1117/12.3112813
3. John M. Bass, Muralidhar M. Balaji, Jeffrey Y. Chen, Gregory Nero, Ignacio A. Rojas, Avion M. Lowery, **Keaton Shurilla**, et al. "Efficient pointing, acquisition, and tracking without external feedback using opposing spiral scanning", Proc. SPIE 14044, Autonomous Systems: Sensors, Processing, and Security for Ground, Air, Sea, and Space Vehicles and Infrastructure 2026, 140440A (11 Jun 2026); https://doi.org/10.1117/12.3095174
2. Schvaneveldt, K. M., Laughlin, A., Guanuna, E., **Shurilla, K.**, Johnson, L., Staker, J., ... & Smalley, D. E. (2022, August). Patterning ITO using a laser cut Kapton® tape mask for flexible PVDF applications. In 2022 IEEE International Flexible Electronics Technology Conference (IFETC) (pp. 1-2). IEEE.
1. Barton, D., Kuttler, R., Weaver, B., Briceno, X., Huffman, B., **Shurilla, K.**, ... & Smalley, D. (2021, July). Photophoretic Optical Trap Display Test Rigs. In 3D Image Acquisition and Display: Technology, Perception and Applications (pp. 3F4A-3). Optica Publishing Group.

Patents

2. (Invention Disclosure to University of Arizona Tech Transfer Office) Acoustoelectrically Enhanced Holographic Video, **Keaton Shurilla** & Matthew Eichenfield
1. SMALLEY, D., LAUGHLIN, E., GUANUNA, E., **SHURILLA, K.**, LUKE, J., SCHVANEVELDT, K., KARAFIN, J. S., & BEVENSEE, B. E. (n.d.). HAPTIC DEVICES (Patent WO/2023/220445).

Teaching Experience

TA for Digital Content Expression and Arts Laboratory **2021 & 2023**
Prof. Yoichi Ochiai & Tatsuki Fushimi

Provided technical assistance to students in their individual media art works. Facilitated and secured the space for a culminating exhibition of the class projects.

Skills & Qualifications

N1 - Japanese Language Proficiency Test （日本語能力試験１級合格） **August 2022**

Service, Outreach, and Other Experiences

- ❑ 2026: Volunteer with Tucson Japanese Language School
- ❑ 2025 Nov: Led hologram workshop for OPTI 527: Diffractive Optics and Holographic Principles
- ❑ 2023 Aug: Presentation on holography to visiting students from Kakogawa high school
- ❑ 2022 Oct: Lab exhibition at the “Miraikan” [video]
- ❑ 2022 Aug: Presentation on ultrasonic phased arrays to visiting students from Kakogawa high school
- ❑ 2018 – 2019: President of the BYU Japanese Student Association
- ❑ 2018: Member of the BYU UX Design Club
- ❑ 2017: Best Buddies volunteer, BYU college chapter
- ❑ 2014 – 2016: Full-Time Volunteer, The Church of Jesus Christ of Latter-Day Saints, Yokohama, Japan
- ❑ 2013: Eagle Scout

Media

- ❑ 2025: Interview on Interrobang Podcast [video]
- ❑ 2023: Research Lab visit on Weekly Ochiai [video]
- ❑ 2022: FUTURES PIXIE DUST RADIO – Interview on Tokyo FM [recording]
- ❑ 2021: “BYU Capstone team creates world-champion educational device” [article]
- ❑ 2021: “Electrical Engineering: A Pathway to New Forms of Creativity” – individual highlight [article]